Aleksei Samoilenko

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Education

Saint Petersburg State University, Specialist Degree in Fundamental Mathematics (BSc+MSc equivalent)
2021 – Present (final year)
Advisor – **Alexander Smirnov**

Research Projects

Iwasawa theory for algebraic tori PDF available

Studied the arithmetic of algebraic tori, primarily norm tori associated with Galois extensions of global fields. Investigated Shafarevich–Tate groups of tori and their behavior in the cyclotomic tower. The class group of a torus is also explored and explicit computations are carried out for tori associated with imaginary quadratic extensions.

Gross and Rohrlich points of infinite order on Jacobians of Fermat curves PDF available

Computed endomorphism algebras of Jacobians of Fermat curves, indicating that the corresponding abelian varieties are not modular in the classical sense. Described noncongruence uniformization of Fermat curves.

Academic Activities

- Active participant in Alexander Smirnov's arithmetic seminar since the beginning of my studies.
- Since summer 2025 I participate in Vasily Golyshev's group; studying Kolyvagin systems and kernels of differential equations.
- Co-organizer (with Aleksei Lvov) of a student seminar on abelian varieties at the faculty, focused on studying the proof of Faltings's theorem and building a local arithmetic geometry community.
- Attended courses on étale cohomology, class field theory, modular forms, Brauer groups, Milnor K-groups and Galois cohomology, oriented cohomology theories and Riemann-Roch theorem, intersection theory, algebraic K-theory, modular representation theory, homological and homotopical algebra, category theory, algebraic geometry and algebraic number theory.
- I have read the following books: Algebraic Number Theory (Cassels, Fröhlich), Algebraic Number Theory (Lang), A Course in Arithmetic (Serre), A Classical Introduction to Modern Number Theory (Ireland, Rosen), Zeta-Functions (Shafarevich), p-adic Numbers, p-adic Analysis, and Zeta-Functions (Koblitz), Introduction to Cyclotomic Fields (Washington), A First Course in Modular Forms (Diamond, Shurman), Algebraic Geometry (Hartshorne), and Homological Algebra (Gelfand, Manin).

Conferences, Workshops, and Summer Schools

Autumn ALGEULER , 2025 , Euler institute (St. Petersburg)

School and Workshop on Explicit Arithmetic Geometry , 2025 , ICTP (Trieste)

IV Conference of Mathematical Centers of Russia , 2024 , PDMI RAS (St. Petersburg)

Student Mathematical School "Algebra and Number Theory" , 2024 ,
HSE , International Laboratory for Mirror Symmetry and Automorphic Forms (Voronovo)

Summer Mathematics School "Algebra and Geometry" , 2023 ,
HSE , Laboratory of Algebraic Geometry and Its Applications (Suzdal)

Teaching Activities

- Teach practical classes in Algebra and Number Theory for computer science students.
- Taught an online Olympiad math circle for secondary school students.